PHP dom Module

JavaScript (in a browser) has the DOM baked-in; PHP has an equivalent API provided by the dom extension, which offers a number of built-in classes for DOM representation. The (new for PHP 8.4) Dom\HTMLDocument class will parse HTML5 and create a data structure representation of our markup, which the following code demonstrates. You can run all of the code in this section from the command line; there is no need to serve via cli-server or equivalent. I'll give you a clear warning when we return to needing a web server SAPI, for example, to work properly.

```
$validHtml = <<<EOS
// ... HTML as per last example
EOS;

$dom = Dom\HTMLDocument::createFromString($validHtml);
var_dump($dom->doctype->name);
var_dump($dom->charset);

Output:
string(4) "html"
string(5) "UTF-8"
```

For versions of PHP prior to 8.4, the DOMDocument class does a similar job but is not aware of HTML5 and parses HTML as version 4, which means the semantic elements introduced by HTML5 will cause Warning errors to be emitted (more on this later). The createFromString() method (and the other factory method createFromFile()) will emit a Warning error if the HTML is malformed (say, a missing closing tag) but does not fully validate the HTML, and even invalid HTML will still result in a DOM structure. Dom\HTMLDocument extends Dom\Document, which is the "modern, spec-compliant equivalent of DOMDocument"; however, the DOMDocument::validate() method is not present on the new class. This is a brand-new feature for PHP 8.4, and hopefully the missing functionality will be added in the future. For now, you'll need to use an external library or service for HTML5 validation.

The Dom\HTMLDocument class exposes the body and head elements as simple read-only properties (remember that head and body are required attributes for valid HTML):

```
// ... $validHtml and $dom as before
var_dump($dom->head->innerHTML);
```